

Market Universe, Data Quality & Account Constraints

GPS Implementation Specification Pack • Prepared for Claude Code • August 28, 2026 • Draft implementation directives; requires securities/compliance counsel review before live personalized recommendations or execution.

Novice eligible universe

Do not implement a fixed “best stock” list. Build a dynamic, rule-based universe. A stock becomes eligible only after it passes all filters, and becomes tradeable only after it additionally passes the signal and account-risk engines.

```
novice_eligible = market_cap_pass AND liquidity_pass AND price_or_fractional_pass AND spread_pass AND
event_risk_pass AND volatility_pass AND data_quality_pass
trade_qualified = novice_eligible AND regime_pass AND confirmation_pass AND target_path_pass AND
account_risk_pass
```

Filter	Initial rule	Notes
Market capitalization	Absolute floor \$10B; preferred core floor \$20B	Do not use \$10M; that admits unsuitable microcaps.
Liquidity	Initial configurable threshold: \$250M average daily dollar volume; core preference \$500M+	Validate threshold against execution data.
Price/accessibility	Prefer \$10–\$125 or confirmed fractional availability	Price alone never qualifies a stock.
Spread	Max spread threshold as % of price and fraction of stop distance	Reject where friction is material.
Events	No Novice new entry ahead of earnings/defined binary events in hold window	Unknown data defaults to block/caution.
Prohibited/conditional	No penny, thin, promotional, leveraged/inverse ETF, low-float, or binary biotech defaults	Advanced products require separate policies.

Small-account mechanics

A \$450 account is a learning-sized account. It can execute valid plans only if position quantity, risk, allocation, and exit mechanics are feasible. Do not lower liquidity/quality standards to make share count look larger.

- Require fractional-share capability for scaled TP1/TP2/runner behavior at small balances, or use an all-in/all-out fallback when fewer than four whole-share-equivalent units can be supported.
- Check settled funds, available buying power, cash versus margin account type, T+1 implications, and broker restrictions before presenting executable sizing.
- Flag account plans whose position allocation is too large even if planned stop risk is within budget.
- Record whether account data is broker-verified, manually entered, delayed, or unavailable.

Data contracts and freshness

Data element	Required source properties	Fail-safe behavior
Quotes/OHLCV	Timestamp, exchange/session, adjusted/unadjusted designation, latency status	No new high-tier signal if stale beyond policy
Corporate actions	Split/dividend adjustments and identifiers	Recalculate historical coordinates; audit changes

Data element	Required source properties	Fail-safe behavior
Earnings/events	Date/time zone, confidence/source, event type	Unknown/ambiguous = no Novice entry or prominent warning
Fundamentals/market cap	As-of date and source consistency	Do not use stale cap to certify eligibility
Account/holdings	Account type, cash, market value, buying power, positions, pending orders	Risk automation disabled if not verified
Broker execution	Capabilities, fractional support, order types, accepted/rejected order events	Use alert-only if capabilities unknown

Market expansion policy

Do not export stock thresholds into other markets. Each asset class requires a separate execution/risk/data engine and independent validation.

Market	Required additional capabilities before release
Options	Underlying-first qualification, strike/expiry, Greeks, IV, bid/ask, open interest, theta, assignment/exercise and spread rules
Futures	Contract roll, point/tick value, margin, sessions, expiry, exchange hours, overnight rules
Forex	Pair quote conversion, pip value, lot sizing, rollover, sessions, macro-event rules
Crypto	Venue/custody, 24/7 data, funding, leverage/liquidation, exchange fragmentation, counterparty controls
Commodities	Contract specs, delivery/roll, report/event calendars, limit-move/seasonality controls